

An Analysis of TensorFlow

TensorFlow

TensorFlow -- Part 1

Google JAX is a machine learning framework for transforming numerical functions. It is described as bringing together a modified version of autograd (automatic obtaining of the gradient function through differentiation of a function) and TensorFlow's XLA (Accelerated Linear Algebra). It is designed to follow the structure and workflow of NumPy as closely as possible and works with TensorFlow as well as other frameworks such as PyTorch. The primary functions of JAX are:

TensorFlow -- Part 2

Google Colab Google also released Collaboratory, a TensorFlow Jupyter notebook environment that does not require any setup. It runs on Google Cloud and allows users free access to GPUs and the ability to store and share notebooks on Google Drive.

TensorFlow -- Part 3

TensorFlow TensorFlow serves as a core platform and library for machine learning. TensorFlow's APIs use Keras to allow users to make their own machine-learning models. In addition to building and training their model, TensorFlow can also help load the data to train the model, and deploy it using TensorFlow Serving. TensorFlow provides a stable Python Application Program Interface (API), as well as APIs without backwards compatibility guarantee for JavaScript, C++, and Java. Third-party language binding packages are also available for C#, Haskell, Julia, MATLAB, Object Pascal, R, Scala, Rust, OCaml, and Crystal. Bindings that are now archived and unsupported include Go and Swift.

TensorFlow -- Part 4

Distribute In both eager and graph executions, TensorFlow provides an API for distributing computation across multiple devices with various distribution strategies. This distributed computing can often speed up the execution of training and evaluating of TensorFlow models and is a common practice in the field of AI.

TensorFlow -- Part 5

Losses To train and assess models, TensorFlow provides a set of loss functions (also known as cost functions). Some popular examples include mean squared error (MSE) and binary cross entropy (BCE).

TensorFlow -- Part 6

Metrics In order to assess the performance of machine learning models, TensorFlow gives API access to commonly used metrics. Examples include various accuracy metrics (binary, categorical, sparse categorical) along with other metrics such as Precision, Recall, and Intersection-over-Union (IoU).

TensorFlow -- Part 7

Edge TPU In July 2018, the Edge TPU was announced. Edge TPU is Google's purpose-built ASIC chip designed to run TensorFlow Lite machine learning (ML) models on small client computing devices such as smartphones known as edge computing.

TensorFlow -- Part 8

Extensions TensorFlow also offers a variety of libraries and extensions to advance and extend the models and methods used. For example, TensorFlow Recommenders and TensorFlow Graphics are libraries for their respective functional. Other add-ons, libraries, and frameworks include TensorFlow Model Optimization, TensorFlow Probability, TensorFlow Quantum, and TensorFlow Decision Forests.

TensorFlow -- Part 9

TensorFlow TensorFlow is Google Brain's second-generation system. Version 1.0.0 was released on February 11, 2017. While the reference implementation runs on single devices, TensorFlow can run on multiple CPUs and GPUs (with optional CUDA and SYCL extensions for general-purpose computing on graphics processing units). TensorFlow is available on 64-bit Linux, macOS, Windows, and mobile computing platforms including Android and iOS. Its flexible architecture allows for easy deployment of computation across a variety of platforms (CPUs, GPUs, TPUs), and from desktops to clusters of servers to mobile and edge devices. TensorFlow computations are expressed as stateful dataflow graphs. The name TensorFlow derives from the operations that such neural networks perform on multidimensional data arrays, which are referred to as tensors. During the Google I/O Conference in June 2016, Jeff Dean stated that 1,500 repositories on GitHub mentioned TensorFlow, of which only 5 were from Google. In March 2018, Google announced TensorFlow.js version 1.0 for machine learning in JavaScript. In Jan 2019, Google announced TensorFlow 2.0. It became officially available in September 2019. In May 2019, Google announced TensorFlow Graphics for deep learning in computer graphics.

TensorFlow -- Part 10

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Retail The e-commerce platform Carousell used TensorFlow to provide personalized recommendations for customers. The cosmetics company ModiFace used TensorFlow to create an augmented reality experience for customers to test various shades of make-up on their face.

TensorFlow -- Part 11

Education InSpace, a virtual learning platform, used TensorFlow to filter out toxic chat messages in classrooms. Liulishuo, an online English learning platform, utilized TensorFlow to create an adaptive curriculum for each student. TensorFlow was used to assess the current abilities of students and also helped decide which content to show based on those capabilities.

TensorFlow -- Part 12

Optimizers TensorFlow offers a set of optimizers for training neural networks, including ADAM, ADAGRAD, and Stochastic Gradient Descent (SGD). When training a model, different optimizers offer different modes of parameter tuning, often affecting a model's convergence and performance.

TensorFlow -- Part 13

TensorFlow is a software library for machine learning and artificial intelligence. It can be used across a range of tasks, but is used mainly for training and inference of neural networks. It is one of the most popular deep learning frameworks, alongside others such as PyTorch. It is free and open-source software released under the Apache License 2.0. It was developed by the Google Brain team for Google's internal use in research and production. The initial version was released under the Apache License 2.0 in 2015. Google released an updated version, TensorFlow 2.0, in September 2019. TensorFlow can be used in a wide variety of programming languages, including Python, JavaScript, C++, and Java, facilitating its use in a range of applications in many sectors.

TensorFlow -- Part 14

AutoDifferentiation AutoDifferentiation is the process of automatically calculating the gradient vector of a model with respect to each of its parameters. With this feature, TensorFlow can automatically compute the gradients for the parameters in a model, which is useful to algorithms such as backpropagation which require gradients to optimize performance. To do so, the framework must keep track of the order of operations done to the input Tensors in a model, and then compute the gradients with respect to the appropriate parameters.

TensorFlow -- Part 15

Social media Twitter implemented TensorFlow to rank tweets by importance for a given user, and changed their platform to show tweets in order of this ranking. Previously, tweets were simply shown in reverse chronological order. The photo sharing app VSCO used TensorFlow to help suggest custom filters for photos.

TensorFlow -- Part 16

TFX TensorFlow Extended (abbrev. TFX) provides numerous components to perform all the operations needed for end-to-end production. Components include loading, validating, and transforming data, tuning, training, and evaluating the machine learning model, and pushing the model itself into production.